

with just mean-reverting short rates. This phenomenon is the *bond excess volatility puzzle* and was documented by the same economist, Robert Shiller (1979), who documented excess volatility in stock markets (See chapter 8).¹⁵ Long-term bond yields also exhibit large excess sensitivity to macroeconomic announcement shocks.¹⁶ Long-term bonds exhibit relatively large volatility and are sensitive to movements in macro and other factors because they exhibit large risk premiums. That is, long-term bonds are very sensitive to factor risk, and to adequately describe factor risk premiums for bonds, we need to explain how factor risk premiums are related across the yield curve.

3.2. MACRO-FACTOR TERM STRUCTURE MODELS

While conventional monetary policy operates on the short end of the yield curve, it influences the long end as well. Bond prices at the long end of the yield curve, however, are (largely but not completely now due to quantitative easing) set by markets, and thus the Fed has much less control of these long-term bonds than over the price of Fed funds. Figure 9.2 shows the three-month T-bill has almost identical movements with the Fed funds rate, which the Taylor rule tells us responds to macro factors. The same macro factors, or the level factor, will affect the entire yield curve through *no arbitrage*.

While prices of two different stocks can diverge wildly because they have different managers and operations, risk-free bonds with different maturities cannot have prices too far out of whack with each other. This is because bonds only differ in terms of their maturity since their cash flows are risk free. Bond markets are chock full of arbitrageurs who pounce on the slightest profitable opportunities when prices of bonds with a particular maturity move out of line with prices of bonds with similar maturities. While there are some limited arbitrage opportunities in bond markets (no one believes—not even ivory-tower academics—that markets are perfectly efficient; see chapter 6), bonds must move in a tight fashion with other bonds. Otherwise, the profit opportunities grow too big, and like in a game of whack-a-mole, smart investors quickly bash their mallets down to remove them. We capture this phenomenon with models where investors receive returns for holding bonds commensurate with their risk, and as a close approximation, we assume bond prices are determined by the *absence* of arbitrage. These are called *term structure models*. The key insight of this class of models is that bond yields are linked to each other—across maturities and across time.

Take, for example, a two-year Treasury note. Investing for a two-year period, we can just go and buy the two-year T-note. We can also invest our money over a

¹⁵Early economic models with *constant* risk premiums were not only unable to explain excess volatilities of long-term bonds, they could not even explain why the yield curve sloped upward, on average (see Backus, Gregory, and Zin (1989) and den Haan (1995)).

¹⁶See Gürkaynak, Sack and Swanson (2005).